

KRISHNA MURARI VUDUGULA

Full-Stack & AI Developer • Computer Science (Data Science) Undergraduate

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PROFESSIONAL SUMMARY

Second-year Computer Science (Data Science) undergraduate with hands-on experience building and deploying full-stack, AI-powered applications using FastAPI, Flask, OpenCV, and SQL-based architectures. Delivered six production-grade projects spanning LLM integration, computer vision, mobile systems, and backend engineering including applications deployed to Vercel and Render with real users. Experienced in API development, database design, and software architecture across the full stack. Seeking internship opportunities in AI/ML, backend systems, or full-stack development to contribute production-ready engineering from day one.

TECHNICAL SKILLS

Languages:	Python, JavaScript, Java, C
Backend:	FastAPI, Flask, REST API Development, JWT Authentication & Authorization, SQLAlchemy, SQLite, MySQL, CRUD Operations, Database Design, API Integration
AI / ML:	Computer Vision, OpenCV, face_recognition, Inference Pipeline, Groq LLM API, Prompt Engineering, Feature Engineering, Scikit-learn, Statistical Modeling
Frontend:	HTML5, CSS3, JavaScript, Bootstrap 5, jQuery, Responsive Design, Mobile-First Development, Frontend Architecture, Client-Server Communication, Dark/Light Theming
Mobile:	Capacitor, Kotlin, Android Development, Background Services
Services:	OpenStreetMap API, Firebase FCM, Asynchronous Programming, Application Deployment
Deployment:	Vercel, Render, GitHub Pages
Tools:	Git, GitHub, Linux CLI, VS Code
CS Core:	Data Structures & Algorithms, OOP, System Design, REST Principles

EDUCATION

B.Tech — Computer Science (Data Science) | Balaji Institute of Technology and Science, Warangal | July 2024 – Present (Expected 2028)

CGPA: 7.88

Relevant coursework: Data Structures & Algorithms, OOP, Database Systems, Machine Learning Foundations, Statistical Modeling, System Design

ACHIEVEMENTS & HIGHLIGHTS

- Shipped **6 production-grade projects** in ~18 months as a first/second-year student, 3 with live deployments (Vercel, Render, GitHub Pages)
- Deployed live SaaS product** — Try Your Prime at try-your-prime.vercel.app, serving real users with live Groq LLM API calls and multi-turn conversation state
- Built enterprise-grade AI attendance system** with 99.60% face recognition accuracy, 61 REST API endpoints, 17 database tables, and native Android integration via Capacitor & Kotlin
- Completed **7 certifications** across ML (Wolfram Research), AI foundations, Python full-stack, and frontend design

PROJECTS

AI Face Attendance Mobile Application — Python · Flask · OpenCV · SQLite · Capacitor · Kotlin · Firebase FCM · OpenStreetMap API [GitHub ↗](#)

Enterprise-grade AI attendance ecosystem combining computer vision, GPS geofencing, real-time tracking, and native Android integration — prepared for institutional deployment.

- Engineered a complete attendance ecosystem** with AI-powered face verification (**99.60% accuracy**), GPS-based geofence validation, automated absent marking, leave management, and attendance analytics dashboards

- Architected and implemented **61 REST API endpoints**, 17 database tables, and 13 frontend interfaces supporting faculty, admin, and institutional workflows — a full system-design exercise at scale
- **Developed native Android background services** with Firebase FCM wake-up notifications, enabling automated check-in/check-out, real-time location tracking, and secure attendance processing via Capacitor & Kotlin
- Implemented **admin dashboards, security logs, duplicate detection, in-app messaging**, and CSV attendance exports supporting large-scale institutional management

Try Your Prime (TYP) — FastAPI · Python · Groq LLM API · Vercel [GitHub ↗](#) [Live ↗](#)

AI-powered conversational self-assessment tool deployed on Vercel — real users, live LLM inference, multi-turn context management.

- **Architected a FastAPI backend** integrating the Groq LLM API; engineered multi-turn prompt engineering logic converting structured user habit inputs into personalized behavioral projections
- Orchestrated full **conversation state management across multi-turn API calls** within a stateless REST architecture — reliable context coherence across sessions
- **Deployed to Vercel** with async streaming UI; application serves **active users** and handles end-to-end LLM inference pipeline from input collection to response rendering

IntelliRoute - Smart Ride Booking — Python · FastAPI · SQL · OpenStreetMap API · Render [GitHub ↗](#) [Live ↗](#)

Full-stack ride-booking and driver-matching platform with traffic insights, route visualization, and role-based dashboards — deployed on Render.

- **Engineered role-based rider and driver dashboards** supporting full ride lifecycle management and real-time status tracking via FastAPI backend and SQL persistence
- Integrated **OpenStreetMap API** for traffic insights and interactive route visualization; implemented client-server communication for real-time ride state updates
- **Deployed to Render** with persistent SQL backend; publicly accessible application handling end-to-end booking flows across multiple user roles

Fee Management System — Python · Flask · SQLAlchemy · SQLite · Bootstrap [GitHub ↗](#)

Secure Flask web application for student fee lifecycle management — payment tracking, due-date alerts, and automated PDF receipt generation.

- **Implemented full CRUD operations** with session-based admin authentication, JWT-ready authorization layer, and parameterized SQL queries via SQLAlchemy to eliminate injection vulnerabilities
- Designed and automated a **due-date alerting system and PDF receipt generation** pipeline, reducing fee verification and administrative overhead significantly

CivicResponse - Civic Issue Reporting Platform — HTML5 · CSS3 · JavaScript · Bootstrap · Flask (in progress) [GitHub ↗](#)

Crowdsourced civic-tech platform for reporting and tracking infrastructure issues with geolocation tagging and status updates.

- Designed a fully **responsive mobile-first interface** supporting issue categorization, geolocation tagging, and real-time status tracking across all device sizes
- Currently developing FastAPI backend with SQL persistence, user authentication, **and municipal analytics dashboards** - active development

Personal Portfolio Website — HTML5 · CSS3 · JavaScript · GitHub Pages [GitHub ↗](#) [Live ↗](#)

- **Designed, implemented, and deployed** a responsive portfolio platform from scratch using CSS Flexbox/Grid, dark/light theme toggle — live on GitHub Pages
- Continuously maintained as a **live proof-of-work**; optimized for Lighthouse performance, linked from all job applications

CERTIFICATIONS

• Machine Learning: Statistical Foundations — Wolfram Research	• AI Foundations: Machine Learning
• ML & Data Science Internship — EduSkills	• AI Internship — CodSoft
• Python Full-Stack Internship — EduSkills	• Frontend Web Design (HTML5/CSS3) — freeCodeCamp
• C Programming — Codetantra	• Foundation Level Code Unnati — Edunet Foundation